Reflective Learning Journal: Digital Development at the Ontario Science Centre

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COOP 2100

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## Section 1: Placement Overview and Learning Objectives

Company Name: Ontario Science Centre
Division: Digital & Innovation Team
Job Title: Co-op Developer
Mentor: Kierstan Hong, Director,
Digital Secondary Supervisor: Joe Cansino, Team Lead, UX

Learning Objectives

To facilitate measurable professional growth during this 12-month placement, I established the following objectives in collaboration with my supervisors:

1. Technical Mastery: Achieve proficiency in core Godot engine functionalities by developing a functional game prototype, with a focus on implementing movement, user interface, and save-state systems.
2. Professional Communication: Enhance oral presentation skills by leading three 10-minute team summaries, which will be evaluated using a feedback rubric assessing clarity and engagement.
3. Collaborative Design: Facilitate 12 monthly "Insight Sessions" with the UI/UX team to bridge the gap between technical development and asset creation.
4. Accessibility Advocacy: Research and implement Web Content Accessibility Guidelines (WCAG) in all game projects, with validation through internal playtest sessions.
5. Interdisciplinary Creative Growth: Complete an introductory music theory course to integrate original audio compositions into Godot projects.

## Section 2: Reflective Journal Entries

Weeks 1–2: Onboarding, Cultural Immersion, and Technical Agility

During the initial phase of my placement, I focused on integrating into the Digital & Innovation Team. I was introduced to my supervisor, Kierstan, and to the broader team of full-time employees and fellow York University co-op students. Our collective mission is to develop educational games that advance the Ontario Science Centre's objective of providing interactive STEM experiences to the public.

Retell: During the first week, we attended formal onboarding sessions where the full-time developers introduced the software stack and IT protocols. Because I was unfamiliar with the Godot engine and GDScript, I utilized the downtime while waiting for hardware to study the engine’s workspace and documentation. By the second week, we received our devices and began working through onboarding packages.

Relate: This period tested my "Agility and Adaptability" competency. Coming from a previous co-op, I recognized the importance of the onboarding phase, but found this environment more collaborative. The morning scrums and consistent office presence facilitated communication that I had lacked in previous roles.

Reflect: I learned that proactive learning is essential when encountering unfamiliar technology. By independently exploring Godot’s structure, I accelerated my learning. I also observed that professional motivation is highly influential. The OSC staff’s commitment to their mission encouraged me to ask technical questions, which facilitated my integration into the digital team more rapidly than I had expected.

Weeks 3–4: Iterative Development and the Accessibility Lens

Retell: My initial technical task involved playtesting and bug-fixing "Salmon Run," a mobile game developed by a previous co-op student. Joey, a fellow co-op developer, and I identified several user interface and game-flow issues, particularly related to stamina tracking and obstacle interactions. I implemented a new energy bar and enhanced enemy visuals to increase their dynamics. In week 4, I transitioned to "Solar Escape," where I addressed accessibility gaps, including keyboard navigation for menus previously reliant on mouse input.

Relate: This work directly supported my objective of cultivating an accessibility-first mindset. Alicia, a full-time developer, introduced the team to Web Content Accessibility Guidelines (WCAG), offering a new perspective on game design.

Reflect: The shift from bug-fixing to accessibility redesign demonstrated that inclusion must be considered from the outset. Retrofitting keyboard navigation into a completed project proved far more challenging than integrating it during initial development. This experience reinforced the importance of comprehensive planning and leveraging external documentation and tutorials to address complex implementation challenges.

Weeks 5–6: Ideation, Presentation, and Strategic Communication.

Retell: In week 5, the team concentrated on brainstorming for the upcoming website launch. I presented two primary concepts: a biome-based "Tower Scaling" game and an "Open World Hub" region. Although the tower concept was well-received, it ultimately evolved into a third proposal, a hero project centred on pollution and climate change, which was selected for development.

Relate: Presenting these concepts enabled me to practice effective oral and written communication. I needed to persuade others of the validity of my ideas while remaining receptive to feedback.

Reflect: I learned that the ideation process is both iterative and non-linear. My initial ideas were not failures but essential steps toward the final Hero Project. Observing the UI/UX co-ops, Jessica and Sepanta demonstrated how art principles inform game design, enabling me to refine my presentations to be more visually and conceptually compelling.

Weeks 7–8: Interdisciplinary Collaboration and Technical Pivot

Retell: In a significant shift, I moved from Godot (2D) to Three.js (3D) to develop a space exploration interactive. This required constant communication with the science team to ensure the copy and scientific concepts were accurately represented. By the end of Week 8, I had a beta version of the scrollable interactive ready for review.

Relate: This period highlighted the importance of curiosity and imagination. Since Three.js was unfamiliar to the entire team, we needed to share our discoveries and collaborate to solve problems.

Reflect: Communicating with the science team was a unique challenge; they are experts in their fields but less familiar with technical programming terms. I had to learn how to translate complex digital concepts into simpler terms to ensure our goals were aligned. This emphasized the significance of empathy in professional communication; understanding the audience's perspective is as vital as the technical execution itself.

Week 9: The Hero Project Kickoff

Retell: We officially kicked off the "Hero Piece" game project, a long-term endeavour that will run through the end of the year. My team, including a writer, science lead, and art designer, decided on a science-fiction approach to climate change.

Relate: Returning to Godot after the transition to Three.js required me to revisit previous documentation and relearn certain processes, demonstrating the agility and adaptability necessary in a multi-project environment.

Reflect: This project kickoff marks the culmination of my first two months of learning. I am now able to apply my accessibility mindset and UI/UX insights from the outset, rather than retrofitting them into an existing build.

## Section 3: Evaluation of Learning Objectives and Growth

Reflecting on the initial two months of my 12-month co-op, I have made substantial progress toward achieving my S.M.A.R.T. goals.

### Technical Proficiency and Agility:

My initial objective was to master the Godot engine. While I have developed a strong understanding of GDScript through work on Salmon Run and Solar Escape, the unexpected transition to Three.js further broadened my technical skill set. I have learned that being a developer at the OSC involves selecting the most appropriate tool for each educational objective. Moving between 2D and 3D environments has reinforced my agility and adaptability.

### Communication and Leadership:

I have successfully led multiple presentations during scrums, achieving my goal of enhancing my oral communication. The selection of my climate change game as a Hero Project provides tangible evidence of my growing ability to persuade others of my ideas. Moving forward, I will continue my Insight Sessions with the UI/UX team to further bridge the gap between development and design.

### Accessibility and Personal Growth:

The most significant change in my professional outlook has been adopting an accessibility-first mindset. Previously, I regarded accessibility as a checklist item; now, I recognize it as an essential component of the creative process, ensuring our games are accessible to the broadest possible audience.

### Conclusion:

This placement has deepened my understanding of the connections between science, education, and technology. I have progressed from a student who follows instructions to a collaborator who contributes original ideas and advocates for user experience. For the remainder of the term, I will concentrate on integrating my developing knowledge of music theory into the Hero Project and on enhancing my ability to communicate technical concepts to non-technical stakeholders.